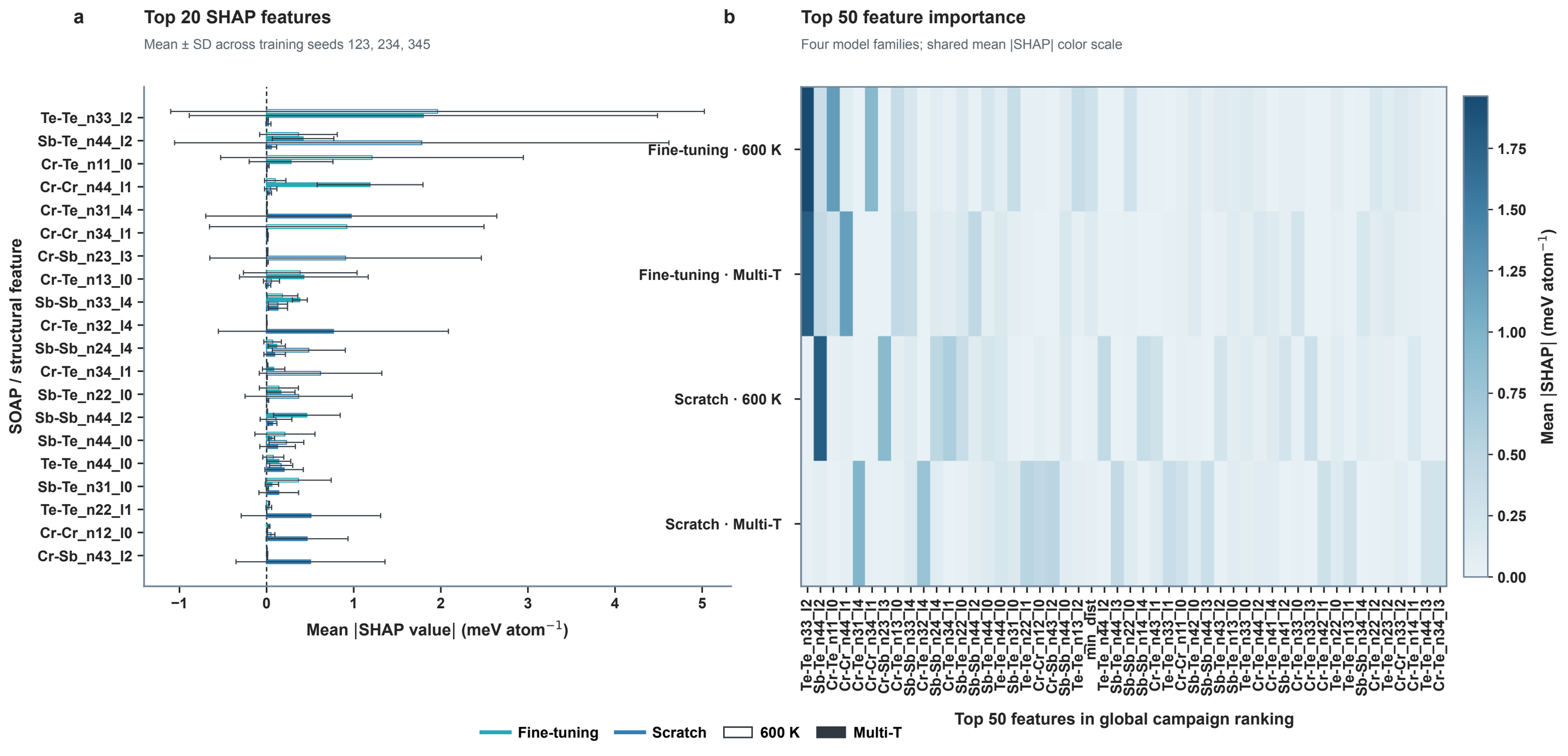

a
Top 20 SHAP features
Mean ± SD across training seeds 123, 234, 345
Te-Te_n33_l2
Sb-Te_n44_l2
Cr-Te_n11_l0
Cr-Cr_n44_l1
Cr-Te_n31_l4
Cr-Cr_n34_l1
Cr-Sb_n23_l3
Cr-Te_n13_l0
Sb-Sb_n33_l4
Cr-Te_n32_l4
Sb-Sb_n24_l4
Cr-Te_n34_l1
Sb-Te_n22_l0
Sb-Sb_n44_l2
Sb-Te_n44_l0
Te-Te_n44_l0
Sb-Te_n31_l0
Te-Te_n22_l1
Cr-Cr_n12_l0
Cr-Sb_n43_l2
SOAP / structural feature
−1
0
1
2
3
4
5
Mean |SHAP value| (meV atom^{-1})
b
Top 50 feature importance
Four model families; shared mean |SHAP| color scale
Fine-tuning · 600 K
Fine-tuning · Multi-T
Scratch · 600 K
Scratch · Multi-T
Te-Te_n33_l2
Sb-Te_n44_l2
Cr-Te_n11_l0
Cr-Cr_n44_l1
Cr-Te_n31_l4
Cr-Cr_n34_l1
Cr-Sb_n23_l3
Cr-Te_n13_l0
Sb-Sb_n33_l4
Cr-Te_n32_l4
Sb-Sb_n24_l4
Cr-Te_n34_l1
Sb-Sb_n44_l2
Sb-Te_n22_l0
Sb-Sb_n44_l0
Te-Te_n44_l0
Sb-Te_n31_l0
Te-Te_n22_l1
Cr-Cr_n12_l0
Cr-Sb_n43_l2
Sb-Sb_n44_l0
Te-Te_n13_l2
min_dist
Te-Te_n44_l2
Sb-Te_n44_l3
Sb-Sb_n22_l0
Sb-Sb_n14_l4
Cr-Te_n43_l1
Te-Te_n33_l1
Cr-Cr_n11_l0
Sb-Te_n42_l0
Sb-Sb_n44_l3
Sb-Te_n43_l2
Sb-Te_n13_l0
Te-Te_n33_l0
Te-Te_n44_l2
Cr-Te_n41_l4
Sb-Te_n41_l2
Cr-Te_n33_l0
Cr-Te_n33_l3
Cr-Te_n42_l1
Te-Te_n22_l0
Te-Te_n13_l1
Sb-Sb_n34_l4
Cr-Te_n22_l2
Te-Te_n23_l2
Cr-Cr_n33_l2
Cr-Te_n14_l1
Te-Te_n44_l3
Cr-Te_n34_l3
Top 50 features in global campaign ranking
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Mean |SHAP| (meV atom^{-1})
Fine-tuning
Scratch
600 K
Multi-T